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KEEPCODING · AI ENGINEERING · FINAL PROJECT

Company Brain

An AI assistant that knows all of a consultancy’s documents and answers in seconds, without any data leaving the laptop

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THE PROBLEM

Consultancies lose hours looking for information they already have

Today

- Information is spread across calls, proposals, contracts and reports
- Finding “what did we agree with this client?” can take hours
- It is confidential: it cannot go to ChatGPT or the cloud

With Company Brain

Ask and get the answer with its source

Summarise a one-hour call in minutes

Draft proposals in your own method



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THE PROJECT'S RULES

Designed for a small business

€0

a month: no cloud, no subscriptions

16 GB

an ordinary laptop (MacBook Air), no servers

2

languages: English and Spanish

77

test documents, invented or public

Not included yet: real client data or "training" the AI to measure. That comes later.

Technical: all local and open source: Ollama · PostgreSQL 18 + pgvector · Docling · Open WebUI · LuLu firewall · launchd services (macOS) · MacBook Air M4 with 16 GB. Corpus: 60 own fictional documents + 17 public ones with commercial licences (CUAD, CORD, UK Statistics Authority and Ararteko reports).



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TRUST AND ETHICS

Four promises the system always keeps

Privacy

Nothing leaves the laptop. The AI has no internet access and backups are encrypted.

Safe test data

Tested only with invented or public documents that allow commercial use.

Respect for authors

No books or protected works are used. Proposals follow my own method of work.

Honesty

Every answer says where it comes from. If it does not know, it says so, and it ignores orders hidden in documents.

Technical: privacy: LuLu blocks internet access for every AI process (checked by “Check health”); backups encrypted with age and signed. Honesty: output guardrail with HHEM-2.1 (scores every claim against its source), a per-language relevance floor that returns “I don’t know”, and per-client isolation. Security: OWASP Top 10:2025 and OWASP Top 10 for LLM Applications 2025 audit.

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THE SYSTEM'S BRAIN

We chose the AI model that fits in a laptop

AI model (free and open)	Fits in the laptop?	Decision
Gemma 4 12B (large)	No: it freezes	Dropped
Qwen 3.5 9B (medium)	No: it freezes	Dropped
Qwen 3.5 4B (small)	Yes, and runs smoothly	Chosen

8 s

to answer a question

45 s

to draft a proposal

3 min

to summarise a 1-hour call

Technical: Qwen3.5-4B (NVFP4 quantisation) in Ollama, 16k-token context, temperature 0.2. Rule: ≤ 14 GB of memory and no critical memory pressure with the search models loaded (own benchmark with vm_stat and memory_pressure). Times: warm median.

HOW IT LEARNS THE BUSINESS

It looks up the documents instead of memorising them

DONE

Clear instructions

Three jobs (answer, summarise, draft) with rules: always cite the source and follow my method.

DONE · THE KEY

Search (RAG)

Before answering, it finds the useful paragraphs in the documents. A new document works at once, with no retraining.

LATER

Training

With few documents of my own, training the AI would cost a lot and add little. We will try it with more data.

Technical: prompt engineering with 3 modes (answer, summarise, draft), mandatory [n] citations and answers in the question's language. Hybrid RAG: pgvector (EmbeddingGemma 2) + PostgreSQL full-text search in ES and EN, RRF fusion (k = 60), 20 candidates → cross-encoder reranker (mMiniLM) → 5 passages. Fine-tuning (QLoRA with Unsloth or MLX) postponed: fewer than 50 own documents and 16 GB.

HOW IT WORKS

From a document in Drive to an answer with its

1 · WHEN A NEW DOCUMENT ARRIVES

Google Drive the company's folder → **Reading** PDF, Word and photos (Docling) → **Passages** indexed by meaning (EmbeddingGemma 2) → **Storage** local database (PostgreSQL with pgvector)

2 · WHEN YOU ASK A QUESTION

Search by meaning and by words (RRF method) → **Filter** keeps the best passages (mMiniLM) → **Writing** the AI answers from them (Qwen 3.5 4B) → **Answer** with sources [1] [2]

Full RAG pipeline, compared before choosing: we tested 3 models that turn text into "meaning" (Qwen3-Embedding 0.6B, bge-m3 and EmbeddingGemma 2). Same quality; EmbeddingGemma 2 was the fastest.

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READY TO USE

A product you use with a double-click

STAGE · INTEGRATION

Easy to use

A chat window like ChatGPT's (Open WebUI), connected to our own local API through a Pipe. Download in Word and PDF.

STAGE · MLOPS (DEPLOYMENT)

Runs itself

Automatic deployment: one button switches it on and off; another checks that all is well.

STAGE · MLOPS (SECURITY)

Secure

Encrypted, signed backup every night and a security review passed (OWASP audit).

Optimised for 16 GB: each part loads only when needed, so the laptop never freezes.

Technical: 4 launchd services (PostgreSQL, Ollama, own API on 127.0.0.1:8765, Open WebUI) · brainctl script (start, stop, health, update, backup) · models loaded on demand.

DOES IT REALLY WORK?

We tested it with 66 questions, in English and

What we asked

- Specific facts: fees, dates, agreements
- Call summaries
- No-answer questions: it must say “I don’t know”
- Documents with traps to trick it

How we checked

- Does it find the right document? (recall@5 and MRR)
- Is the fact exact, with its source?
- A larger AI marks the answers (Gemma 4 12B)
- I rate 20 answers without knowing which is which

22 questions were kept secret until the final test, like an exam nobody has seen before.
Our own offline results panel (instead of Weights & Biases): no data leaves the laptop.

Technical: own metrics (recall@5, MRR, exact fact, citations, refusal) + Ragas (faithfulness and relevancy) with Gemma 4 12B as judge + HHEM. Exam: 22 questions encrypted with age, run once with the frozen configuration (config/selected.toml).



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RESULTS

It finds the source and never invents figures

16/22
questions passed (73%)

94%
finds the right document

22/22
answers with no invented figures

0.80
out of 1: faithful to the sources

What still needs work

Relevance 0.70 out of 1: it sometimes says more than asked. And it fell for both exam traps.
Said “I don’t know” 2 times out of 3 · In practice (44 questions): 41/44

Technical: MRR 0.94 · faithfulness 0.80 (target 0.90) · relevancy 0.70 (target 0.80) · refusal 2/3 (target 90%) · 0 hard blockers · status “pilot, below target”. Report with SHA-256 hashes in reports/final_eval.md.



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DEMO · 4 MINUTES

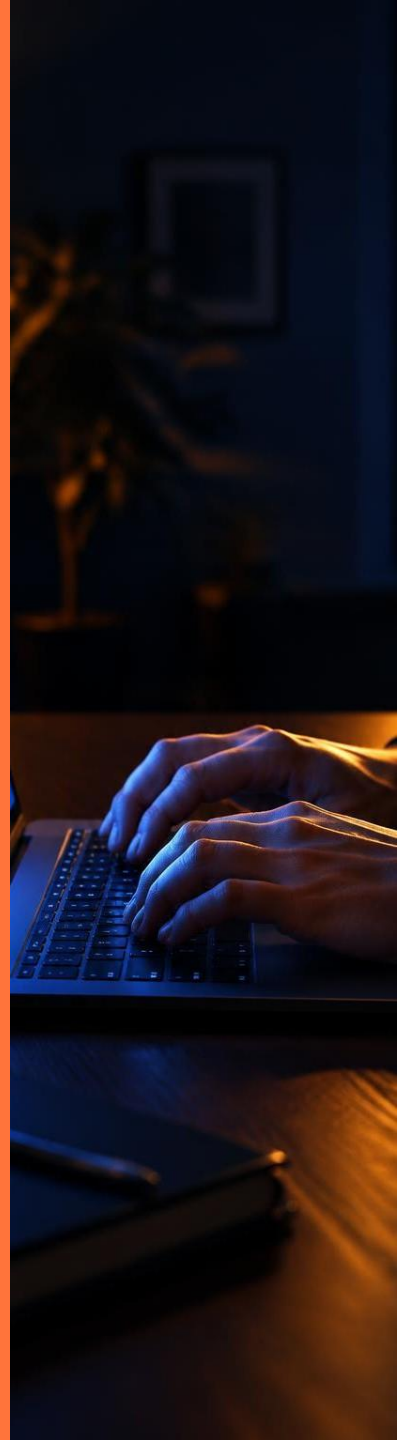
Let’ s see it work

1. A question: “What fee did we agree with Pellmore Dental Group?”, with its source
2. A question with no answer: it says it cannot find it
3. A proposal drafted in my method, downloaded as a PDF

Invented client · the AI works without internet

Technical: Open WebUI → Pipe → local API (127.0.0.1:8765) → hybrid search → reranker → Qwen3.5-4B → answer with sources.

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CONCLUSIONS

A private, useful AI on a laptop, with no monthly cost

What I learned

- You don’t need the biggest AI: a small, well-informed one is enough
- With few documents, searching beats training
- An AI that says “I don’t know” beats an invented answer
- Without your own tests, you can’t tell if it improves

What is annoying

- The firewall asks for permission often, even with the AI off
- You have to choose the mode and the client in every chat
- “Not clearly supported” warnings on correct answers, mostly in Spanish
- The first two are the price of privacy and of never mixing clients

Next steps

- Use it with real clients
- Stronger defences against traps, and try custom training
- A knowledge graph: clients, projects and people
- A quieter firewall: block only the AI
- Suggest the mode and the client, and you confirm
- A warning checker that understands Spanish

Technical: QLoRA (Unsloth or MLX-LM) measured on a new locked exam · knowledge graph on PostgreSQL · prompt-injection defences · recalibrate the English relevance floor · LuLu in passive mode with blocks only for the AI’s processes · intent router with confirmation (one client per chat is deliberate isolation) · the warnings come from HHEM-2.1-Open, trained in English: a correct Spanish claim against the English source scores 0.22 (in English, 0.93) → a multilingual checker, or compare in the source’s language.

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COMPANY BRAIN

Thank you. Questions?

Code: github.com/angelescacosta/company-brain

Test dataset: huggingface.co/datasets/AHAcosta/company-brain-eval

